

## Project Instructions — Stage 02: Tooling Setup

Today's Project Contribution:

Today you'll complete a piece of your full data project.

This task aligns with the **Tooling Setup** lifecycle stage, where you will:

- Set up your project environment and file structure.
- Initialize version control and connect your GitHub repository.
- Create the scaffolding necessary to support future work in your pipeline. By the end of this assignment, your project should include the elements listed below.

### **Deliverable Options (from Stage Scaffold):**

#### **Required:**

- A GitHub repository initialized with the following structure:

```
data/  
data/raw/  
data/processed/  
notebooks/  
src/  
docs/  
reports/  
model/  
README.md
```

- A `.gitignore` file that excludes sensitive files (e.g. `.env`) and system clutter.
- A `README.md` file explaining the project purpose and structure.

#### **Optional (Choose One):**

- A `requirements.txt` or `environment.yml` file for reproducible environment setup.
- A `Makefile` or setup script for automating tasks.
- A `pyproject.toml` or `setup.py` file as preparation for packaging code.

#### **How This Fits Into Your Final Project:**

Your work today builds toward a complete, end-to-end project by establishing a **maintainable and organized technical foundation**. Good structure, version control, and automation will support all future stages of your project lifecycle.

#### **Before next class:**

- Save and organize your files using the designated folder structure.
- Commit and push your initial setup to GitHub.
- Ensure `.gitignore` and `README.md` are included and up to date.

#### **Explicit Chain:**

In your **homework**, you planned your repo setup and connected version control.

**Now, you will adapt that work** to scaffold your live project repo, preparing it to support data ingestion, preprocessing, modeling, and reporting.

**By the end of the course,** your full project will follow the complete lifecycle.